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Optimizing the Performance of Inkjet Printed Green QLED by Precisely Control of Shell Thickness

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Abstract
The quantum dot light-emitting diodes (QLEDs) by inkjet printing are promising candidates for lighting and display applications of the industry. The external quantum efficiency of devices by inkjet printing have gap with that of devices by spin coating, so achieving high efficiency in such inkjet printed devices remains challenging. Herein, we report an inkjet printed green QLED capable of achieving a 21.0% external quantum efficiency.

Author Keywords
Inkjet printing, shell thickness, QLED

1. Introduction
Colloidal quantum dots (QDs) have received extensive attention for their photochemical and photophysical properties such as high photoluminescence quantum yield (PLQY), tunable wavelength, narrow half-width (FWHM), and especially excellent processability.(1–5) In contrast with other traditional luminescent materials, colloidal quantum dots show more suitable for inkjet printing (IJP) technology(6–10). It is a drop-on-demand printing technology and has many advantages such as maskless patterns and low material consumption(11). Therefore, it is particularly important to study the combination of quantum dot technology and IJP technology to produce IJP-QLED that can meet the needs of the display industry in the future.

However, the low efficiency and stability of IJP-QLED hinder its development and application. This is mainly due to Auger recombination (AR), Foster resonance energy transfer (FRET) and quenching of QD(12). In addition, the limitation of IJP technology on quantum dot solvents (as quantum dot inks) also causes a loss of efficiency and stability(11,13–15). Therefore, many strategies have been carried out to solve these problems, including adjusting the structure and ligands of the internal quantum dots(16), developing suitable ink systems(13,17,18), and new post-processing(19). Recently, some research groups have proposed various ligands to passivate the traps on the surface of quantum dots to improve charge injection, thereby obtaining higher device performance(16). The red IJP-QLED using this strategy shows an EQE of more than 16%, and the half-brightness life can reach millions of hours(16). In addition, the shell layer of the quantum dot also plays an important role in the performance of the device, because the number and thickness of the shell layer have an influence on the band gap, quenching and FRET of the quantum dot(20–25). At the same time, it proves that the double shell can significantly improve the performance of the device.

Above of all, as a precise micron scale preparation method, IJP and IJP QLED are easily affected by many factors, such as the materials (QD, solvents, carrier transporting materials) and processing method (film formation, bank structure and so on). Although the development of IJP QLED of has made great progress, most of the works focused on red quantum dots, because their structure is relatively simple to be analyzed. As for the green QD, due to their quaternary alloyed core structure and multi-shells of ZnS, the boundary between core and shell is not clear. The development of green QLED is hence lagged behind.

Herein, we optimized the green IJP QLED by exquisite control of QD shell thickness. The device demonstrated the max EQE of 21%, which is the record of IJP green QLED in literature.

2. Result and discussion

Fig. 1 shows the transmission electron microscopy (TEM) images of the QDs with different thickness outer shell. As Fig.1(a) showing, the QD with two different structure core/shell (C/S) and core/shell/shell(C/S/S) were synthesized with diameters of 10 nm and 12.1 nm. Correspondingly, the thickness of second shells is 2.1 nm (~7 ZnS monolayers).

The relative devices were prepared and their performance are shown in Fig. 2. The device with C/S/S exhibit a higher current density and luminance, which indicate the effective carrier recombination, as shown in Fig. 3. As for the C/S QD, more electrons is expected to inject, but without enough holes to recombine, leading to less photons generated. This study verified a very subtle change in the green quantum dot shell can have a very big impact on the performance of the whole device.

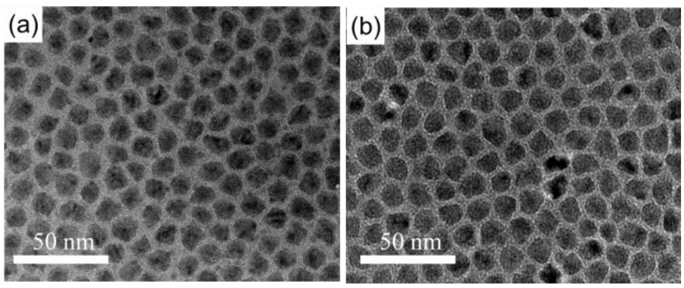

Fig. 1. The TEM images of green QD with diameters of ~10 nm(C/S) and 12.1 nm (C/S/S).

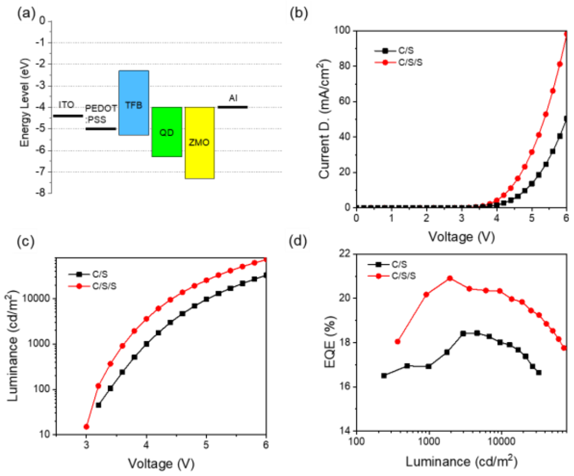

Fig. 2. The device performance of IJP green QLED.

To evaluate the suitability of as-prepared QDs in devices by inkjet printing and the importance of their second shell thickness, the QLEDs were fabricated with a typical normal device structure (Fig. 2) by inkjet printing. The PEDOT:PSS, TFB, QDs, and ZnMgO nanoparticles were deposited on top of indium-tin oxide (ITO). Finally, aluminum was evaporated on top as cathode. The relative devices were prepared and their performance are shown in Fig. 2. As for the C/S QD, more electrons are expected to inject, but without enough holes to recombine, leading to less photons generated. This study verified a very subtle change in the green quantum dot shell can have a very big impact on the performance of the whole device. Through current-voltage-luminance measurement, QLEDs with different thickness shell were characterized. As shown in Fig. 2, C/S/S device shows the higher current density, indicating that the suitable thickness shell can adjust the energy band to improve the injection of carriers. Besides, the C-QD device has the lowest current density. This can be explained that the over thick shell for C-QD increases the traps of its surface, due to the large stress of the thick shell growth and large specific surface area of QD. Fig. 2 shows that the turn-on voltage of C/S and C/S/S devices are the about 3.4 V and 3.2 V @100 cd/m^2, respectively. Meanwhile, the C/S/S device shows the higher luminance up to 72700 cd/m^2, compared to C/S devices.

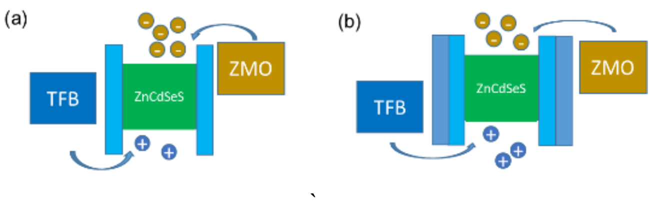

Fig. 3. The mechanism of carrier recombination in QLED for different QD structures.

The device with C/S/S exhibit a higher current density and luminance, which indicate the effective carrier recombination, as shown in Fig. 3. As for the C/S QD, more electrons are expected to inject, but without enough holes to recombine, leading to less photons generated. This study verified a very subtle change in the green quantum dot shell can have a very big impact on the performance of the whole device. Through current-voltage-luminance measurement, QLEDs with different thickness shell were characterized. That indicates this thickness of QDs benefited in the charge injection and recombination, due to the optimal bandgap. As shown in Figure 3, the C/S/S device shows the best balanced current. The balanced current and outstanding film PLQY results that the B-QD device shows the highest electroluminance (@6 V) and EQE (21 %), compared to C/S devices. As we all know, this is the first time that green QLED by inkjet printing achieves over 20% so far.

3. Conclusion

By optimizing the thickness of shell, we synthesized a good green QD. Based on that, the green QLED was fabricated by inkjet printing can achieve a EQE of over 21 % and luminance up to 72700 cd/m^2.

4. Acknowledgment

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5. References

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